Upgrade

Astra Automation

NetApp
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Upgrade

Upgrade overview

The Upgrade App API provides CRUD (Create-Read-Update-Delete) access to software components that are eligible for upgrade, and to initiate upgrades. The availability of upgrades is determined based on packages registered with the Packages API.

List all upgrades

GET `/accounts/{account_id}/core/v1/upgrades`

Returns a JSON array of all matching resources in the collection. If individual fields are specified in the request, the items JSON array will contain the fields requested for each matching resource in the order specified.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
include	string	query	False	Indicates which fields should be returned when listing a collection. URL examples: "include=id", "include=id,name" OpenAPI examples: "id" • example: id,componentName,upgradeVersion

Name	Type	In	Required	Description
limit	string	query	False	Indicates the maximum number of resources to return when listing a collection. URL examples: "limit=2" OpenAPI examples: "2"
filter	string	query	False	Indicates which fields should be matched in order to return a resource when listing a collection. Supported operators are: • eq - Equality • lt/gt - Less than/Greater than • lte/gte - Less than or equals to/Greater than or equals to URL example: "filter=field%20eq%20%27value%27" OpenAPI examples: "field eq 'value'"

Response

```
Status: 200, Returns a JSON array of all matching resources in the
collection. If individual fields are specified in the request, the items
JSON array will contain the fields requested for each matching resource in
the order specified.
```

Name	Type	Required	Description
items	array[Any of: upgrade_1.1_get_response_body, array[]]	True	
metadata	type_astra_metadata_list	True	

Example response

```
{
  "type": "application/astra-upgrades",
  "version": "1.1",
  "items": [
    [
      "01982783-b1eb-4dca-a3fe-a385a3186c53",
      "acc",
      "21.07.1"
    ],
    [
      "0a5abab2-39b2-4101-87b9-0d9b8f537ca1",
      "acc",
      "21.07.2"
    ]
  ],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```

Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	False	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	False	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	

Name	Type	Required	Description
detail	string	False	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	False	
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't found.",
  "status": "404"
}
```

Definitions

See Definitions

type_astra_stateDetails

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata_update

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Name	Type	Required	Description
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

upgrade_1.1_get_response_body

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-upgrade" • example: application/astra-upgrade

Name	Type	Required	Description
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • example: 1.1
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema • example: aa9a8e88-c012-55b1-b514-7cd94dc79008
componentName	string	True	JSON string containing the name of the type of software that can be upgraded. Defined values are: • "acc" • "acs" • "trident" • "kubernetes" • example: trident

Name	Type	Required	Description
componentInstance	string	True	JSON string containing the URI of the component that can be upgraded. Defined values are: • Minimum length: 3 • Maximum length: 4095 • example: <a href="https://<dns_name>/accounts/0b311ae7-d89a-4a11-a52c-1349ca090415/topology/v1/clouds/fdda3ff3-a46a-43a4-902e-444fde2baeba/storageBackends/72d19c3c-eb43-4bec-b23e-a228c900aded</dns_name>" class="bare">https://<dns_name>/accounts/0b311ae7-d89a-4a11-a52c-1349ca090415/topology/v1/clouds/fdda3ff3-a46a-43a4-902e-444fde2baeba/storageBackends/72d19c3c-eb43-4bec-b23e-a228c900aded</dns_name></a>
componentID	string	True	JSON string containing a globally unique identifier of the component that can be upgraded. Defined values are: • Conforms to the Astra Identifier Schema • example: 72d19c3c-eb43-4bec-b23e-a228c900aded

Name	Type	Required	Description
upgradeVersion	string	True	JSON string containing the SemVer of the software image file that can be upgraded to. Defined values are: • Conforms to the SemVer Schema • example: 21.07.1
currentVersion	string	True	JSON string containing the current SemVer of the entity to be upgraded. Defined values are: • Conforms to the SemVer Schema • example: 21.04.1
dependencies	array[string]	True	JSON array of JSON strings, each containing the UUID of an upgrade that must be completed before this upgrade can proceed. Used to determine which upgrades will be performed as prerequisites and in which order. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
state	string	True	JSON string containing a value indicating the operational state of the upgrade. If auto-upgrade is enabled, upgrades will start in the "scheduled" state. Otherwise, upgrades will start in the "proposed" state. Defined values are: • "unavailable" - The upgrade is not available at this time • "proposed" - The upgrade is proposed but will not automatically run until approved • "scheduled" - The upgrade is approved but is not ready to be installed due to dependencies, time windows, or available resources • "running" - The upgrade is in progress • "complete" - The upgrade has completed • "failed" - The upgrade did not complete • example: scheduled

Name	Type	Required	Description
stateDesired	string	True	JSON string containing a value indicating the desired state for "state". This field is only included when the user can directly request a change in the state of the resource. Defined values are: • "proposed" - The upgrade is not approved and must not be performed • "scheduled" - The upgrade is approved and must be performed in the specified time window • "running" - The upgrade must be performed immediately, regardless of the time window • example: scheduled
stateDetails	array[type_astra_stateDetails]	True	JSON array of JSON objects that lists additional details about the current "state". If no details are available, this array will be empty. • Minimum Items: 0

Name	Type	Required	Description
metadata	type_astra_metadata_update	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

type_astra_metadata_list

Name	Type	Required	Description
continue	string	False	
count	number	False	
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

Retrieve an upgrade

GET `/accounts/{account_id}/core/v1/upgrades/{upgrade_id}`

Returns the stored Upgrade API resource in the JSON response body.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
upgrade_id	string	path	True	ID of the upgrade collection to list

Response

```
Status: 200, Returns the stored Upgrade API resource in the JSON response
body.
```

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-upgrade" • example: application/astra-upgrade
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • example: 1.1
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema • example: aa9a8e88-c012-55b1-b514-7cd94dc79008

Name	Type	Required	Description
componentName	string	True	JSON string containing the name of the type of software that can be upgraded. Defined values are: • "acc" • "acs" • "trident" • "kubernetes" • example: trident
componentInstance	string	True	JSON string containing the URI of the component that can be upgraded. Defined values are: • Minimum length: 3 • Maximum length: 4095 • example: <a href="https://<dns_name>/accounts/0b311ae7-d89a-4a11-a52c-1349ca090415/topology/v1/clouds/fdda3ff3-a46a-43a4-902e-444fde2baeba/storageBackends/72d19c3c-eb43-4bec-b23e-a228c900aded</dns_name>" class="bare">https://<dns_name>/accounts/0b311ae7-d89a-4a11-a52c-1349ca090415/topology/v1/clouds/fdda3ff3-a46a-43a4-902e-444fde2baeba/storageBackends/72d19c3c-eb43-4bec-b23e-a228c900aded</dns_name></a>

Name	Type	Required	Description
componentID	string	True	JSON string containing a globally unique identifier of the component that can be upgraded. Defined values are: • Conforms to the Astra Identifier Schema • example: 72d19c3c-eb43-4bec-b23e-a228c900aded
upgradeVersion	string	True	JSON string containing the SemVer of the software image file that can be upgraded to. Defined values are: • Conforms to the SemVer Schema • example: 21.07.1
currentVersion	string	True	JSON string containing the current SemVer of the entity to be upgraded. Defined values are: • Conforms to the SemVer Schema • example: 21.04.1
dependencies	array[string]	True	JSON array of JSON strings, each containing the UUID of an upgrade that must be completed before this upgrade can proceed. Used to determine which upgrades will be performed as prerequisites and in which order. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
state	string	True	JSON string containing a value indicating the operational state of the upgrade. If auto-upgrade is enabled, upgrades will start in the "scheduled" state. Otherwise, upgrades will start in the "proposed" state. Defined values are: • "unavailable" - The upgrade is not available at this time • "proposed" - The upgrade is proposed but will not automatically run until approved • "scheduled" - The upgrade is approved but is not ready to be installed due to dependencies, time windows, or available resources • "running" - The upgrade is in progress • "complete" - The upgrade has completed • "failed" - The upgrade did not complete • example: scheduled

Name	Type	Required	Description
stateDesired	string	True	JSON string containing a value indicating the desired state for "state". This field is only included when the user can directly request a change in the state of the resource. Defined values are: • "proposed" - The upgrade is not approved and must not be performed • "scheduled" - The upgrade is approved and must be performed in the specified time window • "running" - The upgrade must be performed immediately, regardless of the time window • example: scheduled
stateDetails	array[type_astra_stateDetails]	True	JSON array of JSON objects that lists additional details about the current "state". If no details are available, this array will be empty. • Minimum Items: 0

Name	Type	Required	Description
metadata	type_astra_metadata_update	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Example response

```
{
  "type": "application/astra-upgrade",
  "version": "1.1",
  "id": "aa9a8e88-c012-55b1-b514-7cd94dc79008",
  "componentName": "trident",
  "componentInstance": "https://<dns_name>/accounts/0b311ae7-d89a-4a11-
a52c-1349ca090415/topology/v1/clouds/fdda3ff3-a46a-43a4-902e-
444fde2baeba/storageBackends/72d19c3c-eb43-4bec-b23e-a228c900aded",
  "componentID": "72d19c3c-eb43-4bec-b23e-a228c900aded",
  "upgradeVersion": "21.07.1",
  "currentVersion": "21.04.1",
  "dependencies": [],
  "state": "scheduled",
  "stateDesired": "scheduled",
  "stateDetails": [],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```

Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	False	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	False	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	False	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	False	
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't found.",
  "status": "404"
}
```

Definitions

See Definitions

type_astra_stateDetails

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata_update

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Name	Type	Required	Description
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

Modify an upgrade

PUT `/accounts/{account_id}/core/v1/upgrades/{upgrade_id}`

Replaces the entire stored resource with the supplied JSON request body while preserving key values that are

not user-modifiable.

Side Effects

- Upgrades are scheduled or run based on desired state.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
upgrade_id	string	path	True	ID of the upgrade collection to list

Request Body

Replaces the entire stored resource with the supplied JSON request body while preserving key values that are not user-modifiable.

Side Effects

- Upgrades are scheduled or run based on desired state.

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-upgrade" • example: application/astra-upgrade
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • example: 1.1

Name	Type	Required	Description
id	string	False	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema • example: aa9a8e88-c012-55b1-b514-7cd94dc79008
componentName	string	False	JSON string containing the name of the type of software that can be upgraded. Defined values are: • "acc" • "acs" • "trident" • "kubernetes" • example: trident

Name	Type	Required	Description
componentInstance	string	False	JSON string containing the URI of the component that can be upgraded. Defined values are: • Minimum length: 3 • Maximum length: 4095 • example: <a href="https://<dns_name>/accounts/0b311ae7-d89a-4a11-a52c-1349ca090415/topology/v1/clouds/fdda3ff3-a46a-43a4-902e-444fde2baeba/storageBackends/72d19c3c-eb43-4bec-b23e-a228c900aded</dns_name>" class="bare">https://<dns_name>/accounts/0b311ae7-d89a-4a11-a52c-1349ca090415/topology/v1/clouds/fdda3ff3-a46a-43a4-902e-444fde2baeba/storageBackends/72d19c3c-eb43-4bec-b23e-a228c900aded</dns_name></a>
componentID	string	False	JSON string containing a globally unique identifier of the component that can be upgraded. Defined values are: • Conforms to the Astra Identifier Schema • example: 72d19c3c-eb43-4bec-b23e-a228c900aded

Name	Type	Required	Description
upgradeVersion	string	False	JSON string containing the SemVer of the software image file that can be upgraded to. Defined values are: • Conforms to the SemVer Schema • example: 21.07.1
currentVersion	string	False	JSON string containing the current SemVer of the entity to be upgraded. Defined values are: • Conforms to the SemVer Schema • example: 21.04.1
dependencies	array[string]	False	JSON array of JSON strings, each containing the UUID of an upgrade that must be completed before this upgrade can proceed. Used to determine which upgrades will be performed as prerequisites and in which order. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
state	string	False	JSON string containing a value indicating the operational state of the upgrade. If auto-upgrade is enabled, upgrades will start in the "scheduled" state. Otherwise, upgrades will start in the "proposed" state. Defined values are: • "unavailable" - The upgrade is not available at this time • "proposed" - The upgrade is proposed but will not automatically run until approved • "scheduled" - The upgrade is approved but is not ready to be installed due to dependencies, time windows, or available resources • "running" - The upgrade is in progress • "complete" - The upgrade has completed • "failed" - The upgrade did not complete • example: scheduled

Name	Type	Required	Description
stateDesired	string	False	JSON string containing a value indicating the desired state for "state". This field is only included when the user can directly request a change in the state of the resource. Defined values are: • "proposed" - The upgrade is not approved and must not be performed • "scheduled" - The upgrade is approved and must be performed in the specified time window • "running" - The upgrade must be performed immediately, regardless of the time window • example: scheduled
stateDetails	array[type_astra_stateDetails]	False	JSON array of JSON objects that lists additional details about the current "state". If no details are available, this array will be empty. • Minimum Items: 0

Name	Type	Required	Description
metadata	type_astra_metadata_update	False	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Example request

```
{
  "type": "application/astra-upgrade",
  "version": "1.1",
  "stateDesired": "running"
}
```

Response

```
Status: 204, Indicates if the upgrade resource was updated.
```

Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	False	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	False	
status	string	True	
correlationID	string	False	
invalidFields	array[invalidFields]	False	List of invalid request body fields

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 409, Conflict
```

Name	Type	Required	Description
type	string	True	
title	string	True	

Name	Type	Required	Description
detail	string	False	
status	string	True	
correlationID	string	False	
invalidFields	array[invalidFields]	False	List of invalid request body fields

Example response

```
{
  "type": "https://astra.netapp.io/problems/10",
  "title": "JSON resource conflict",
  "detail": "The request body JSON contains a field that conflicts with an idempotent value.",
  "status": "409"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	False	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	False	
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/1",
  "title": "Resource not found",
  "detail": "The resource specified in the request URI wasn't found.",
  "status": "404"
}
```

Definitions

See Definitions

type_astra_stateDetails

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata_update

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

Name	Type	Required	Description
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

invalidFields

Name	Type	Required	Description
name	string	True	Name of the invalid request body field

Name	Type	Required	Description
reason	string	True	Reason why the request body field is invalid

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